ME403 – Introduction to Robotics

Lab 00: Introduction Week Handout

Sabanci University | Spring 2025-26

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Lab Outcomes

By the end of this lab, you will be able to:

- Configure a Python environment for robotic control using Mamba or venv.
- Establish and verify communication with the Dobot Magician over USB-serial.
- Understand the relationship between relative and absolute joint angles.
- Manually calculate and verify Forward Kinematics (FK) for a 4-axis arm.

1. Overview

This lab session focuses on setting up your development environment and verifying the connection with the **Dobot Magician** (USB-serial). Run the scripts directly from `magician/`.

Week 1 Note

MG400 (Ethernet/TCP-IP) exercises are covered in the Week 1 handout (`lab01`). See `Students/01_SecondWeek/` for the MG400 student scripts.

Included Scripts

Script	Purpose
Dobot Magician (`Students/00_IntroductionWeek/magician/`)	
`01_init_check.py`	Verifies USB connection, clears alarms, and moves to safe ready pose. Run first.
`02_joint_control.py`	Interactive REPL: enter joint angles in degrees to move joints.
`03_relative_joint_control.py`	Body-frame FK exercise: enter relative joint angles, observe the conversion chain.

2. Quick Start Guide

2.1. Dobot Magician (USB-serial)

Magician — Mamba (Recommended)

```
# 1. Activate environment and install
mamba activate dobot
pip install -r Students/00_IntroductionWeek/magician/requirements.txt

# 2. Verify connection
cd Students/00_IntroductionWeek/magician
python 01_init_check.py
```

Magician — Python venv

```
python3 -m venv .venv           # Linux/macOS
python -m venv .venv            # Windows PowerShell
source .venv/bin/activate       # Linux/macOS
# .venv\Scripts\Activate.ps1    # Windows PowerShell
pip install -r Students/00_IntroductionWeek/magician/requirements.txt
cd Students/00_IntroductionWeek/magician
python 01_init_check.py
```

3. Hardware Configuration

3.1. Dobot Magician (USB-serial)

1. **Surface:** Place on a flat, stable surface. Clear a 40 cm radius around the base.
2. **Posture:** Manually set the arm to a neutral posture (both segments at ≈45°) before powering on.
3. **Power:** Connect the 12 V DC adapter. **USB power alone is insufficient.**
4. **Communication:** Connect the USB cable (CP210x USB-to-UART bridge).

Status Signals:

- **Green LED + Single Beep:** Ready.
- **Red LED + Rapid Beeping:** Alarm state — run `01_init_check.py` to clear.
- **Yellow/Flashing LED:** Busy (homing or executing a command).

4. Software and OS Setup

4.1. Linux Setup (Dobot Magician only)

Users must have permission to access serial devices. Run the following once:

```
sudo usermod -a -G dialout $USER
# Log out and log back in for changes to take effect.
```

The Magician typically appears as `/dev/ttyUSB0` or `/dev/ttyACM0`.

4.2. Windows Setup

The driver is usually handled by Windows Update. To find the port:

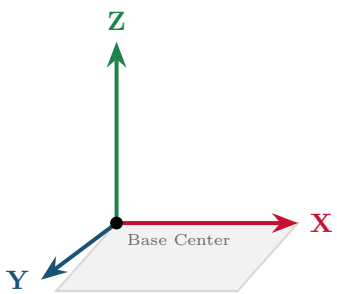
1. Open **Device Manager**.
2. Expand **Ports (COM & LPT)**.
3. Look for **Silicon Labs CP210x USB to UART Bridge** (e.g., `COM3`).

4.3. macOS Setup

Modern macOS versions include the driver. The device will appear as: `/dev/cu.usbserial-*` or `/dev/cu.SLAB_USBtoUART`.

5. Dobot Magician Reference (USB-serial)

5.1. Coordinate System (Cartesian)



- **X-axis:** Forward (pointing away from base).
- **Y-axis:** Left/Right (Left is positive).
- **Z-axis:** Vertical (Up is positive).
- **Units:** Millimeters (mm).
- **Rotation (R):** Tool rotation in degrees (°).

5.2. Joint Definitions and Safe Limits

Joint	Motion	Safe Range	Units
J1	Base Rotation	−90 to 90°	deg
J2	Rear Arm Elevation	0 to 85°	deg
J3	Forearm Elevation	−10 to 85°	deg
J4	End-Effector Rotation	−90 to 90°	deg

Cartesian Dimension	Safe Operating Range	Unit
X (Reach)	120 to 315 mm	mm
Y (Slew)	±158 mm	mm
Z (Height)	5 to 155 mm	mm

6. Forward Kinematics Exercise (Script 03)

Script `03_relative_joint_control.py` introduces the concept of **body-frame (relative) joint angles** and makes the FK transformation chain visible in the terminal output.

6.1. What Are Relative Joint Angles?

Body-Frame vs. World-Frame Angles

Each joint angle can be measured from two different reference directions:

Joint	Relative (body-frame)	Absolute (world-frame)
J1	Base rotation from world X-axis	Same as relative
J2	Shoulder elevation from horizontal	Same as relative
J3	Elbow offset *from the upper arm*	$= \mathrm{J2} + \mathrm{J3}_{\mathrm{rel}}$
J4	Wrist offset *from the forearm*	$= \mathrm{J3}_{\mathrm{abs}} + \mathrm{J4}_{\mathrm{rel}}$

J1 and J2 are identical in both conventions. J3 and J4 are where they differ: a relative J3 of 10° means the forearm is 10° above the upper arm, regardless of how high the shoulder is.

6.2. Conversion Chain

The script applies this chain before every move:

$$j_{3,\mathrm{abs}} = j_{2,\mathrm{rel}} + j_{3,\mathrm{rel}}$$
$$j_{4,\mathrm{abs}} = j_{3,\mathrm{abs}} + j_{4,\mathrm{rel}} = j_{2,\mathrm{rel}} + j_{3,\mathrm{rel}} + j_{4,\mathrm{rel}}$$

Firmware note: the Dobot Magician uses a parallel linkage and its firmware already interprets the J3 channel as a body-frame offset (so $j_{3,\mathrm{fw}} = j_{3,\mathrm{rel}}$ trivially). Only J4 needs the absolute wrist angle: $j_{4,\mathrm{fw}} = j_{4,\mathrm{abs}}$.

6.3. Worked Example (Dobot Magician)

Input: `0 20 10 0` (all relative)

Step	Value
$j_{3,\mathrm{abs}}$	$= 20° + 10° = 30°$
$j_{4,\mathrm{abs}}$	$= 30° + 0° = 30°$
$j_{3,\mathrm{fw}}$	$= 10°$ (trivial for Magician)
$j_{4,\mathrm{fw}}$	$= 30°$ $(= j_{4,\mathrm{abs}})$
FK reach	$= 135\cos(20°) + 147\cos(30°) \approx 254\,\mathrm{mm}$
FK height	$= 135\sin(20°) + 147\sin(30°) \approx 120\,\mathrm{mm}$

Safety Note

Joint bounds are always applied to the **firmware angles** (absolute), not the relative inputs. Entering $j_{3,\text{rel}} = 10°$ when $j_{2,\text{rel}} = 80°$ would produce $j_{3,\text{abs}} = 90°$, which exceeds the J3 firmware bound of 85° and will be clamped with a warning before the move executes.

6.4. Usage and Terminal Output

```
cd Students/00_IntroductionWeek/magician
python 03_relative_joint_control.py
```

Sample terminal output for input `0 20 10 0`:

```
Abs: J1=  0.00  J2= 20.00  J3= 30.00  J4= 30.00
Rel: J1=  0.00  J2= 20.00  J3= 10.00  J4=  0.00
> 0 20 10 0

  Relative:  J1_rel=   0.00  J2_rel=  20.00  J3_rel=  10.00  J4_rel=   0.00
  Absolute:  j1_abs=   0.00  j2_abs=  20.00  j3_abs=  30.00  j4_abs=  30.00
  Firmware:  j1_fw=    0.00  j2_fw=   20.00  j3_fw=   10.00  j4_fw=   30.00
  [FK]       X= 254.17  Y=   0.00  Z= 119.72  R= 30.00
```

Challenge: before running the script, calculate the expected X, Y, Z by hand using the FK formulas. Then compare your answer with the `[FK]` line in the terminal.

7. Troubleshooting

7.1. Dobot Magician

Magician: Common Connection Issues

Symptom	Recommended Action
`Access Denied`	Ensure **DobotStudio** and other Python scripts are closed.
`Device not found`	Verify both USB and Power cables. Check `ls /dev/tty*` (Unix).
Motors don't move	Check the 12 V supply. If light is red, run `01_init_check.py`.
Erratic movement	Reset to neutral posture and power cycle the robot.

8. Safety Mandates

Mandatory Safety Rules

1. **Workspace:** Maintain a clean area. No cables in the reach zone (40 cm radius around the base).
2. **Initialization:** Always run `01_init_check.py` before starting your own scripts.
3. **Helper Functions:** Use `safe_move()` from `utils.py`. It enforces software limits.

4. **Supervision:** Stay within reach of the power cable at all times.